

# Swapnil S Mane

Visiting Faculty @ IIT Madras Zanzibar | Ph.D. @ IIT Jodhpur | PMRF Fellow

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| CONTACT INFORMATION | Office F9<br>IIT Madras Zanzibar<br>Tanzania, P73P+RW<br>✉ E-mail: mswapnil108.sm@gmail.com                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Homepage: <a href="https://swapnilsmane.github.io">swapnilsmane.github.io</a><br>Google Scholar: <a href="#">SwapnilSMane</a><br>Linkedin: <a href="#">swapnil-s-mane-168567136</a><br>YouTube: <a href="#">@swapnilsmane-iitj</a> |
| RESEARCH INTERESTS  | Learning on Graphs · Network Science · Natural Language Processing · Social Networks · AI for Social Good                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                    |
| EDUCATION           | <p><b>Ph.D.</b>, Computer Science and Engineering 2021–2026 (30<sup>th</sup> April)<br/>Indian Institute of Technology Jodhpur<br/>Advisors: Dr. Suman Kundu and Prof. Rajesh Sharma<br/>Thesis: A Socio-Computational Framework for Modeling User Aggression in Social Media: Dataset, Detection and Forecasting. CGPA: 8.60/10</p> <p><b>M.Tech.</b>, Computer Engineering 2019–2021<br/>Advisors: Dr. Vaibhav Khatavkar<br/>College of Engineering Pune Technological University, Pune CGPA: 8.60/10</p> <p><b>B.Tech.</b>, Computer Science and Engineering 2015–2019<br/>Rajarambapu Institute of Technology, Ishwarpur</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                    |
| EXPERIENCE          | <p><b>Visiting Faculty</b>, Data Science and Artificial Intelligence, IIT Madras Zanzibar Oct 2026 – Present<br/><b>Postdoctoral Researcher</b> at Wadhvani School of Data Science &amp; Artificial Intelligence, IIT Madras Feb 2026 – Sep 2026<br/><i>Advised by:</i> Dr. Suman Kundu</p> <p><b>Data Scientist</b> in Keydabra Inc., Atlanta, U.S. April 2021–July 2021<br/><i>Advised by:</i> Dr. Srinivas Kilambi, Ph.D. IIT Madras<br/>Developed predictive analyses to infer behavioral trends and support data-driven decision processes.</p> <p><b>Data Scientist</b> in OneDot Enterprises Pvt. Ltd., Pune, India June 2020–Aug. 2020<br/>Political assessment modeling</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                    |
| PUBLICATIONS        | <p>CONFERENCE</p> <p><b>Swapnil Mane</b>, Suman Kundu, and Rajesh Sharma. <a href="#">INDICAG: An Explainable Agentic Framework for Indic-Multilingual Multidimensional Aggression Detection</a> In Proceedings of the ACM Web Conference 2026, pp. 9361-9372. (<b>WWW 2026</b>).</p> <p><b>Swapnil Mane</b>, Suman Kundu, and Rajesh Sharma. <a href="#">TSGAN: Temporal Social Graph Attention Network for Aggressive Behavior Forecasting</a>. In <i>Proceedings of the AAAI Conference on Artificial Intelligence</i>, vol. 39, no. 27, pp. 28249-28257 (<b>AAAI 2025</b>).</p> <p>Vidit Garg, <b>Swapnil Mane</b>, and Suman Kundu. <a href="#">MACO-Sync: Multi-agent Ant Colony Optimization for Synchronized Arrival Coordination</a>. In International Conference on Applied Algorithms, pp. 403-415. Cham: Springer Nature Switzerland, 2026.</p> <p><b>Swapnil Mane</b> and Vaibhav Khatavkar. <a href="#">CascadeNS: Confidence-Cascaded Neurosymbolic Model for Sarcasm Detection</a>. In Proceedings of the International Conference on Recent Advances in Intelligent and Sustainable Technologies (p. 59). Springer Nature, 2026.</p> <p>Vaibhav Khatavkar, <b>Swapnil Mane</b>, and Parag Kulkarni. <a href="#">Event Uncertainty for Twitter Data Using Thematic Context Vector</a>. In <i>NIELIT's International Conference on Communication, Electronics and Digital Technologies</i>, pp. 135-146. Singapore: Springer Nature Singapore, 2024.</p> <p><b>Swapnil Mane</b> and Vaibhav Khatavkar. <a href="#">Effective feature extraction for intrusion detection systems using non-negative matrix factorization and univariate analysis</a>. <i>SACAIM</i>, 2021.</p> |                                                                                                                                                                                                                                    |

## JOURNALS

**Swapnil Mane**, Suman Kundu, and Rajesh Sharma. You are what your feeds make you: A study of user aggressive behavior on Twitter. *Applied Intelligence* 55, 385, 2025.

**Swapnil Mane**, Suman Kundu, and Rajesh Sharma. A Survey on Online Aggression: Content Detection and Behavioral Analysis on Social Media. *ACM Computing Surveys* 57, no. 7, pp: 1-36, 2025.

**Swapnil Mane** and Vaibhav Khataavkar. Researchers eye-view of sarcasm detection in social media textual content. *International Journal of Creative Research Thoughts (IJCRT)*. Vol. 9, Issue 8, pp. 150-157, 2021.

## RESEARCH GRANTS

*GYĀN-SETU : Multilingual Graph Representation Frameworks to Elicit, Model, Estimate, Diagnose and Explain Learners' Prior Knowledge in India's Higher Education* 2026

ANRF Advanced Research Grant (ARG)

PI: Dr. Suman Kundu; Co-PI: Prof. Mitesh Khapra, WSAI, IIT Madras

Role: Contributed to problem formulation, technical framework design, and preparation of the proposal narrative, and methodology.

## MENTORING

Mentored 2 M.Tech and 7 B.Tech students (IIT Jodhpur / IIT Madras) 2022–2026.  
Selected outcomes:

Garge et al. *MACO-Sync: Multi-agent Ant Colony Optimization for Synchronized Arrival Coordination*, Intl. Conf. on Applied Algorithms, 2026.

Co-mentored work under review for KDD 2027.

## TEACHING ASSISTANT EXPERIENCE

**NOC5\_CS19 Data Mining, NPTEL** Spring 2025

Instructor: Prof. Pabitra Mitra, IIT Kharagpur

**NOC24\_CS120 Social Networks, NPTEL** Fall 2024

Instructor: Prof. Sudarshan Iyengar, IIT Ropar

CSL7870 Learning on Graphs and its Applications Fall 2025

Instructor: Dr. Suman Kundu, IIT Jodhpur

CSL4050 Data Visualization Spring 2025

Instructor: Dr. Suman Kundu, IIT Jodhpur

CSL4030 Data Engineering Fall 2024

Instructor: Dr. Romi Banerjee, IIT Jodhpur

CSL7640 Natural Language Understanding Spring 2024

Instructor: Prof. Lipika Dey, IIT Jodhpur

CSL7670 Machine Learning Fundamentals Fall 2023

Instructor: Dr. Anand Mishra, IIT Jodhpur

CSL1010 Introduction to Computer Science Spring 2023

Instructor: Dr. Palash Das, IIT Jodhpur

CSL1010 Introduction to Computer Science Fall 2022

Instructor: Dr. Dip Sankar Banerjee, IIT Jodhpur

CSL6010 Cyber Security Spring 2022

Instructor: Dr. Debasis Das, IIT Jodhpur

CSL3010 Design and Analysis of Algorithms Fall 2021

Instructor: Dr. Pallavi Jain, IIT Jodhpur

CT-20008 Principles of Programming Languages Spring 2021

Instructor: Dr. Vaibhav Khataavkar, COEP, Pune

## CERTIFICATIONS

Applied Natural Language Processing, **NPTEL** 2021

Deep Learning, **NPTEL** 2021

Data Science Professional, **Coursera** 2018

Natural Language Processing with Python, **Udemy** 2018

Python for Data Science and Machine Learning Bootcamp, **Udemy** 2018

Prepare Manuscript, **Elsevier** 2018

Python (Basic), **HackerRank** 2018

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
|                         | Data Science using Python, <b>MITU Skillologies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2018                                                                                   |
|                         | Programming, Data Structure And Algorithms Using Python, <b>NPTEL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2018                                                                                   |
|                         | High-Performance Computing, <b>IIT Bombay</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2017                                                                                   |
| INVITED TALKS           | Research talk titled “From Detection to Forecasting: A Socio-Computational Framework for Modeling User Aggression in Social Media” at College of Engineering Pune Technological University (COEP), Maharashtra                                                                                                                                                                                                                                                                                                                                            | 2026                                                                                   |
|                         | Lectures on Foundational Machine Learning at <a href="#">Government Colleges of Engineering Karad, Ratnagiri, and Avasari Khurd</a> , Maharashtra                                                                                                                                                                                                                                                                                                                                                                                                         | 2024                                                                                   |
|                         | Talk on Language Meets Intelligence: Real-World Applications of NLP at <a href="#">Government Polytechnic College, Jodhpur</a>                                                                                                                                                                                                                                                                                                                                                                                                                            | 2023                                                                                   |
|                         | Sessions on Information Technology at <a href="#">Kendriya Vidyalaya, IIT Jodhpur</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2023                                                                                   |
|                         | Talk on Basics of Machine Learning at Pimpri Chinchwad Polytechnic College Pune                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2021                                                                                   |
| RELEVANT COURSEWORK     | Machine Learning, Deep Learning, Natural Language Understanding, Social Network Analysis, Advanced Database                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                        |
| COURSE PROJECTS         | Multilingual ngram-based convolutional network for aspect category detection in online reviews<br><i>Advised by:</i> <a href="#">Prof. Lipika Dey</a>                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">🔗</a> <a href="#">📄</a>                                                    |
|                         | Knowledge Distillation: Aggression Detection Using Mini BERT<br><i>Advised by:</i> <a href="#">Prof. Mayank Vatsa</a>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">🔗</a> <a href="#">📄</a>                                                    |
|                         | KnowldgeNet Relationship Classification Using a Pre-trained Model<br><i>Advised by:</i> <a href="#">Prof. Lipika Dey</a>                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">🔗</a> <a href="#">📄</a>                                                    |
|                         | Analysis of Author’s Abusive Behaviour Community on Twitter<br><i>Advised by:</i> <a href="#">Dr. Suman Kundu</a>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">🔗</a> <a href="#">📄</a>                                                    |
|                         | Contextual Journal Recommendation and Query Search Engine<br><i>Advised by:</i> <a href="#">Prof. Gaurav Harit</a>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <a href="#">🔗</a> <a href="#">📄</a>                                                    |
| ACTIVITIES & SERVICES   | Reviewer KDD, NeurIPS, LoG 2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026                                                                                   |
|                         | Reviewer Journal of Big Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026                                                                                   |
|                         | Reviewer Journal of Ambient Intelligence and Humanized Computing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026                                                                                   |
|                         | Reviewer Discover Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026                                                                                   |
|                         | Reviewer Social Network Analysis and Mining                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025                                                                                   |
|                         | Volunteer, ACM IKDD International Conference on Data Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2024                                                                                   |
|                         | Member, Institution of Engineering and Technology (IET), Pune                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2019–2022                                                                              |
|                         | Member, Indian Society for Technical Education (ISTE), Pune                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2016–2020                                                                              |
|                         | Student Cultural Representative, RIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2017–2019                                                                              |
| AWARDS AND SCHOLARSHIPS | <ul style="list-style-type: none"> <li>• Travel Grant for poster presentation at NetSciX Conference</li> <li>• Outstanding <b>Teaching Assistant Award</b>, CSE, IIT Jodhpur</li> <li>• Got selected for “<b>Prime Minister Research Fellowship</b>” (PMRF, GoI)</li> <li>• Department <b>Excellence Award</b>, CSE, RIT</li> <li>• Institute <b>Best Student of the Year Award</b>, RIT</li> <li>• 1<sup>st</sup> in coding competition, CODE-IT, PVPIT, Sangli</li> <li>• Drama Best Director Award, Purushottam Karandak, Kolhapur division</li> </ul> | Feb. 2025<br>May 2023<br>Oct. 2022<br>Feb. 2019<br>Apr. 2019<br>July 2018<br>Oct. 2018 |
| REFERENCES              | <b>Dr. Suman Kundu</b> ( <a href="mailto:suman@dsai.iitm.ac.in">suman@dsai.iitm.ac.in</a> )<br>Assistant Professor, DS & AI, IIT Madras<br><b>Dr. Rajesh Sharma</b> ( <a href="mailto:rajesh.sharma@plaksha.edu.in">rajesh.sharma@plaksha.edu.in</a> )<br>Professor, CS & AI, University of Tartu, Estonia & Plaksha University<br><b>Dr. Lipika Dey</b> ( <a href="mailto:lipikadey@gmail.com">lipikadey@gmail.com</a> )<br>Professor, CSE, Ashoka University & Former Chief Scientist at TCS Research                                                   |                                                                                        |